Master Manuscript V4: Structural Consistency and Residue Sanity

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1 The Residue Sanity Gate: Metric-Volumetric Invariance

To ensure structural consistency in the EABC framework, we establish the Residue Sanity Gate. For any full-dimensional Euclidean lattice $\Lambda \subset \mathbb{R}^d$, the radial Epstein-Zeta function is defined as:

$$Z_\Lambda^{\text{len}}(s) = \sum_{0 \neq x \in \Lambda} \|x\|^{-s}.$$

The fundamental geometric consistency condition at the pole $s = d$ is:

$$\boxed{\operatorname*{Res}_{s=d} Z_\Lambda^{\text{len}}(s) = \frac{2\pi^{d/2}}{\Gamma(d/2)\,\operatorname{covol}(\Lambda)}} \tag{1}$$

Crucial Clarification on Invariance: The residue is *not* a topological invariant, but a *metric-volumetric lattice invariant*. Under dilation $\Lambda \mapsto c\Lambda$, the zeta function transforms as $Z_{c\Lambda}^{\text{len}}(s) = c^{-s} Z_\Lambda^{\text{len}}(s)$, yielding:

$$\operatorname*{Res}_{s=d} Z_{c\Lambda}^{\text{len}} = c^{-d} \operatorname*{Res}_{s=d} Z_\Lambda^{\text{len}}.$$

More generally, for any $A \in GL(d, \mathbb{R})$, we have $\operatorname{covol}(A\Lambda) = |\det A| \operatorname{covol}(\Lambda)$, leading to:

$$\boxed{\operatorname*{Res}_{s=d} Z_{A\Lambda}^{\text{len}} = |\det A|^{-1} \operatorname*{Res}_{s=d} Z_\Lambda^{\text{len}}.}$$

Thus, the residue is invariant under all volume-preserving linear transformations ($|\det A| = 1$), including orientation-reversing ones.

2 Divisor Information and Winding Audit (Argument Principle)

For a meromorphic function, the winding number along a closed contour γ in the resolvent set is given by the argument principle:

$$\nu_\gamma = \frac{1}{2\pi i} \oint_\gamma \frac{Z'(s)}{Z(s)}\, ds = N_Z(\gamma) - N_P(\gamma) \in \mathbb{Z}.$$

This integer is invariant under homotopies of the contour within the complement of the divisor. This forms the clean dichotomy:

$$\begin{aligned} \operatorname{Res}_{s=d} Z_\Lambda &\quad\longleftrightarrow\quad \text{metric/volumetric pole strength,} \\ \frac{1}{2\pi i} \oint_\gamma d\log Z_\Lambda &\quad\longleftrightarrow\quad \text{discrete divisor / winding information.} \end{aligned}$$

3 Comparative Divisor and Winding Audit for D_4 and E_8

Because D_4 lives in dimension 4 and E_8 in dimension 8, there is no canonical continuous cross-dimensional transition along which a single winding number could be transported directly from D_4 to E_8. Instead, we perform a rigorous comparative divisor audit by analyzing the divisors $\operatorname{div} Z^{\text{len}}_{D_4}$ and $\operatorname{div} Z^{\text{len}}_{E_8}$ using concentric contours centered at the respective dimensional poles:

$$\gamma_{D_4}(R) : |s-4| = R, \quad \gamma_{E_8}(R) : |s-8| = R.$$

The audit maps the discrete step functions:

$$R \mapsto \big(\nu_{D_4}(R), \nu_{E_8}(R)\big),$$

designed to determine the radial distances at which zeros and secondary poles appear and to identify cancellations among factor singularities. No cross-dimensional transport law is assumed.

4 Monomial Fourier–Projection Uniqueness Theorem

Define the $L = 1$ projection functional

$$\mathcal{P}[f] := 2\int_0^1 (1-s) f(s)\, ds$$

and, for integers $a \geq 0$ and $m \geq 1$, the monomial Fourier family

$$f_{a,m}(s) := s^a \sin(2\pi m s).$$

Set $k = 2\pi m$ and

$$M_a(k) := \int_0^1 s^a \sin(ks)\, ds.$$

For $a \geq 2$, two integrations by parts, together with $\sin k = 0$ and $\cos k = 1$, give

$$M_a(k) = -\frac{1}{k} - \frac{a(a-1)}{k^2} M_{a-2}(k), \tag{2}$$

with initial values $M_0(k) = 0$ and $M_1(k) = -1/k$.

Introduce

$$x := \frac{1}{k^2}, \qquad J_a(x) := kM_a(k).$$

Then

$$J_0 = 0, \qquad J_1 = -1,$$

and

$$J_a(x) = -1 - a(a-1)xJ_{a-2}(x), \qquad a \geq 2. \tag{3}$$

Hence $J_a \in \mathbb{Z}[x] \subset \mathbb{Q}[x]$ for every a.

Since

$$\mathcal{P}[f_{a,m}] = 2(M_a - M_{a+1}),$$

define

$$D_a(x) := J_a(x) - J_{a+1}(x).$$

Then

$$\mathcal{P}[f_{a,m}] = \frac{2}{k} D_a\left(\frac{1}{4\pi^2 m^2}\right). \tag{4}$$

Theorem 4.1 (Monomial Fourier–Projection Uniqueness)**.** *For every $a \in \mathbb{N}_0$ and $m \in \mathbb{N}$,*

$$\boxed{\mathcal{P}[s^a \sin(2\pi m s)] = 0 \iff a = 1.}$$

Proof. For $a = 1$, the recursion gives $J_1 = J_2 = -1$, hence $D_1 \equiv 0$. For $a = 0$, $D_0 = 1$. For $a = 2$, one finds $D_2(x) = -6x \not\equiv 0$. For every $a \geq 3$, the recursion yields

$$J_a(x) = -1 + a(a-1)x + O(x^2), \qquad J_{a+1}(x) = -1 + a(a+1)x + O(x^2),$$

so

$$D_a(x) = -2ax + O(x^2),$$

and therefore D_a is a nonzero polynomial in $\mathbb{Q}[x]$.

For fixed $m \geq 1$,

$$x_m = \frac{1}{4\pi^2 m^2}$$

is transcendental because π is transcendental. A nonzero polynomial with rational coefficients has only algebraic roots. Consequently $D_a(x_m) = 0$ can occur only when $D_a \equiv 0$, which happens exactly for $a = 1$. □

The cancellation at $a = 1$ is nontrivial at the level of moments:

$$M_1(k) = M_2(k) = -\frac{1}{k} = -\frac{1}{2\pi m} \neq 0,$$

while

$$\mathcal{P}[s \sin(2\pi m s)] = 2(M_1 - M_2) = 0.$$

Thus the linear exponent is the unique kernel exponent within this monomial Fourier family. This statement is algebraic-analytic; no physical vacuum interpretation is implied by the theorem itself.

4.5.2 Disjoint Logarithmic Shell Audit: Support Weights and Detailed Balance

To separate genuine bulk behavior from smoothing caused by cumulative averages, we repeat the support-weight and reversibility audit on disjoint prime-value shells. Let $\widehat{\pi}^{\mathrm{shell}} = (\widehat{\pi}_2, \widehat{\pi}_3, \widehat{\pi}_4)$ denote the empirical frequencies of the three coarse-grained support sectors S_2, S_3, S_4 in a shell, and let

$$\varpi = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$$

be the predicted reference weight vector. For the shell transition matrix T^{shell}, define the detailed-balance defect by

$$D_{\mathrm{DB}}^{\mathrm{shell}} := \left\| \mathrm{diag}(\varpi) T^{\mathrm{shell}} - \left(T^{\mathrm{shell}}\right)^{T} \mathrm{diag}(\varpi) \right\|_{F}. \tag{5}$$

Thus $D_{\mathrm{DB}}^{\mathrm{shell}} = 0$ is the exact detailed-balance condition for the chosen reference weights, whereas a nonzero value measures the finite-shell defect in Frobenius norm.

Table 1: Disjoint logarithmic shell audit for primes up to $p = 500{,}000$. The reported values are finite-range numerical data. They provide evidence for stabilization of the support frequencies and a reduced detailed-balance defect in the largest shell, but do not by themselves prove an asymptotic limit.

Shell	Prime-value range p	Count	$\widehat{\pi}^{\mathrm{shell}}(S_2, S_3, S_4)$	$D_{\mathrm{DB}}^{\mathrm{shell}}$	Diagonal entries (T_{22}, T_{33}, T_{44})
1	[100, 1,000)	143	(0.4825, 0.2587, 0.2587)	0.035057	(0.3529, 0.0541, 0.1622)
2	[1,000, 5,000)	501	(0.4970, 0.2475, 0.2555)	0.008821	(0.3735, 0.1371, 0.1181)
3	[5,000, 20,000)	1,593	(0.4997, 0.2530, 0.2473)	0.034353	(0.4138, 0.1241, 0.1269)
4	[20,000, 100,000)	7,330	(0.4991, 0.2508, 0.2502)	0.008211	(0.4155, 0.1447, 0.1468)
5	[100,000, 500,000)	31,946	(0.4992, 0.2507, 0.2501)	**0.006970**	(0.4225, 0.1587, 0.1576)

Support-weight stabilization. The disjoint-shell frequencies approach the reference vector closely over the tested range. In the largest shell,

$$\widehat{\pi}^{\mathrm{shell}} = (0.4992, 0.2507, 0.2501),$$

so the componentwise deviations from ϖ are

$$(-8 \times 10^{-4},\ 7 \times 10^{-4},\ 1 \times 10^{-4}).$$

This is strong finite-range shell evidence that the observed support weights are stable near the predicted values. It is not, on its own, a proof that $\widehat{\pi}^{\mathrm{shell}} \to \varpi$ as $p \to \infty$. Such an asymptotic statement requires an independent analytic argument, including an exact description of the pushforward from the underlying residue classes to the three support sectors.

Detailed-balance defect. The sequence

$$0.035057,\ 0.008821,\ 0.034353,\ 0.008211,\ 0.006970$$

is manifestly non-monotone. Nevertheless, the largest shell attains the smallest reported defect. The appropriate finite-data conclusion is therefore an overall endpoint reduction accompanied by substantial shell-to-shell fluctuations. This is compatible with finite-size arithmetic correlations, but the present audit does not identify Hardy–Littlewood correlations as the unique cause and does not establish

$$\lim_{p \to \infty} D_{\mathrm{DB}}^{\mathrm{shell}} = 0.$$

Diagonal persistence. The diagonal entries are reported without qualitative dynamical labels. In the largest shell one finds

$$T_{22} = 0.4225, \qquad T_{33} = 0.1587, \qquad T_{44} = 0.1576.$$

Their differing magnitudes are an empirical property of the implemented coarse-grained transition matrix. Any combinatorial explanation in terms of preimage multiplicities or cross-transition counts must be established separately from the numerical table.

Epistemic status. The shell audit strengthens the numerical evidence because the intervals are disjoint and therefore do not inherit the deterministic smoothing of cumulative averages. The current status is

Support-weight convergence	STRONG FINITE RANGE EVIDENCE,
Detailed-balance defect	ENDPOINT REDUCTION WITH FLUCTUATIONS,
Asymptotic detailed balance	NOT YET PROVED,
Borel-measure generation	NOT CLAIMED.

The terminology "Borel weights" is therefore avoided here: ϖ is treated as a predicted support-weight vector (or, once proved, as a Dirichlet-pushforward weight), while the standard Borel measure-theoretic framework is presupposed rather than derived from the discrete arithmetic model.

5 Residue Specializations for E_8 and D_4

For the standard E_8 lattice, $\mathrm{covol}(E_8) = 1$, and therefore

$$\boxed{\operatorname*{Res}_{s=8} Z^{\mathrm{len}}_{E_8}(s) = \frac{\pi^4}{3}.}$$

For the standard root lattice

$$D_4 = \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2\},$$

we have $\mathrm{covol}(D_4) = 2$, hence

$$\boxed{\operatorname*{Res}_{s=4} Z^{\mathrm{len}}_{D_4}(s) = \pi^2.}$$

If D_4 is rescaled to covolume 1, the corresponding residue is $2\pi^2$. These values are normalization-sensitive consequences of the general residue formula and constitute mandatory consistency checks for any proposed product or Dirichlet-series representation.

5.1 Status of the comparative winding stairs

The comparative winding functions are defined by

$$\nu_{D_4}(R) = \frac{1}{2\pi i} \oint_{|s-4|=R} d\log Z^{\mathrm{len}}_{D_4}(s), \qquad \nu_{E_8}(R) = \frac{1}{2\pi i} \oint_{|s-8|=R} d\log Z^{\mathrm{len}}_{E_8}(s).$$

Their step locations are determined by the complete divisors of the corresponding meromorphic continuations. A proposed factorization may be used for this purpose only after it has passed both the pole-structure check and the Residue Sanity Gate. Accordingly, the comparative staircase is defined here as an audit object; a full D_4 staircase is not claimed until an independently validated D_4 factorization or divisor description is supplied.

6 Claim-Wall

- **Metric-Volumetric Residue Formula:** PROVEN
- E_8 **specialization** ($\mathrm{covol} = 1$) **and standard** D_4 **specialization** ($\mathrm{covol} = 2$)**:** PROVEN
- **Argument Principle / Winding Invariance:** PROVEN
- **Konkrete Produktdarstellung:** VALIDATED (only if pole structure matches the Gate)
- **Comparative Divisor Audit (**D_4 **vs** E_8**):** DEFINED / AUDIT PENDING

- **Monomial Fourier–Projection Uniqueness Theorem:** PROVEN
- **Disjoint-shell support-weight stabilization:** STRONG FINITE-RANGE EVIDENCE
- **Detailed-balance shell defect:** ENDPOINT REDUCTION WITH FLUCTUATIONS
- **Asymptotic detailed balance:** NOT YET PROVED
- **Generation of Borel measure theory from the discrete model:** NOT CLAIMED
- **Physical / anyonic interpretation of divisor or winding data:** HEURISTIC / NOT CLAIMED

11.6.1 Quaternionic Three-Dimensional Contraction of the Fano Rotational Term

Let ϕ_{ijk} denote the totally antisymmetric Fano three-form associated with an oriented octonionic multiplication table. On any chosen quaternionic subalgebra $\mathrm{span}\{1, e_a, e_b, e_c\} \cong \mathbb{H}$, the restriction of ϕ to the imaginary three-plane is, after choosing a standard Levi–Civita tensor ϵ, of the form

$$\phi_{ijk}\big|_{\{a,b,c\}} = \sigma\, \epsilon_{ijk}, \qquad \sigma \in \{+1, -1\}. \tag{6}$$

The sign σ records the relative orientation of the chosen Fano line and the standard orientation of the physical three-space. In particular, for the orientation convention used in the present audit, the Fano line $(1, 3, 2)$ gives

$$\phi_{123} = -1 = -\epsilon_{123}, \tag{7}$$

so that $\sigma = -1$ on the selected $\{1, 2, 3\}$ sector.

Let the antisymmetric part of the three-dimensional velocity gradient be

$$\Omega_{jk} = \frac{1}{2}(\partial_j u_k - \partial_k u_j) = -\frac{1}{2}\epsilon_{jkl}\omega_l, \tag{8}$$

with the displayed sign fixing the vorticity convention. Define the restricted Fano rotational contraction by

$$\mathcal{R}_F^{(3)}(\omega) := \sum_{i,j,k=1}^{3} \phi_{ijk}\, \omega_i \Omega_{jk}. \tag{9}$$

Theorem 6.1 (Orientation-dependent quaternionic contraction)**.** *Under the above conventions,*

$$\boxed{\mathcal{R}_F^{(3)}(\omega) = -\sigma\, |\omega|^2.} \tag{10}$$

Hence the contraction is negative definite only when the Fano orientation and the spatial orientation are aligned ($\sigma = +1$). For the project convention $(1, 3, 2)$, one has $\sigma = -1$ and therefore

$$\mathcal{R}_F^{(3)}(\omega) = +|\omega|^2. \tag{11}$$

Proof. Using $\phi_{ijk} = \sigma\epsilon_{ijk}$ and the three-dimensional vorticity identity,

$$\mathcal{R}_F^{(3)} = -\frac{\sigma}{2} \sum_{i,j,k,l=1}^{3} \epsilon_{ijk}\epsilon_{jkl}\, \omega_i\omega_l \tag{12}$$

$$= -\frac{\sigma}{2} \sum_{i,l=1}^{3} 2\delta_{il}\, \omega_i\omega_l \tag{13}$$

$$= -\sigma \sum_{i=1}^{3} \omega_i^2. \tag{14}$$

This proves the identity exactly. □

The preceding theorem is an algebraic contraction identity. It does not, by itself, provide a coercive estimate for the classical Navier–Stokes vortex stretching term. Two separate issues remain.

First, changing the relative orientation σ changes the sign of the contraction. Consequently, negative definiteness is not an orientation-free consequence of the Fano algebra. A dissipative interpretation therefore requires an independent dynamical or geometric principle that fixes the physical embedding, the orientation, and the coupling sign.

Second, the classical three-dimensional enstrophy production is

$$\mathcal{S}_{\mathrm{NS}} = \int_{\mathbb{R}^3} \omega_i S_{ij} \omega_j \, dx, \tag{15}$$

where S is the symmetric strain tensor. The Fano contraction above is a separate algebraic observable unless one proves that it enters the actual Navier–Stokes evolution with a fixed coefficient and controls $\mathcal{S}_{\mathrm{NS}}$ in a scaling-critical norm. No such coercive bridge is established by the contraction identity alone.

Gate status. The precise status of the sixth Fano gate is therefore

$G_{\mathrm{NS}}^{(6a)}$	EXACT PASS $\quad (\phi_{ijk}S_{jk} = 0)$,
$G_{\mathrm{NS}}^{(6b)}$ algebraic reduction	EXACT PASS,
$G_{\mathrm{NS}}^{(6b)}$ 3D contraction	$\mathcal{R}_F^{(3)} = -\sigma\|\omega\|^2$,
orientation-independent damping	NOT ESTABLISHED,
$G_{\mathrm{NS}}^{(6c)}$ coercive bridge	OPEN,
global 3D regularity	NOT ESTABLISHED.

11.6.4 Morrey Threshold and the Discrete $W^{1,6}$ Direction-Coherence Gate

The Fano-angle identity on a selected quaternionic three-plane gives an exact orientation-sensitive representation of the angle between two nonzero vorticity directions. Writing

$$\xi(x,t) = \frac{\omega(x,t)}{|\omega(x,t)|}$$

wherever $\omega \neq 0$, and using $\phi|_{\mathbb{H}_{\mathrm{Im}}} = \sigma\epsilon$ with $\sigma \in \{\pm 1\}$, one has

$$\boxed{\|\phi(\xi(x,t),\xi(y,t),\cdot)\| = \|\xi(x,t)\times\xi(y,t)\| = |\sin\theta(x,y,t)|.} \tag{16}$$

This identity is exact, but by itself it supplies no spatial modulus of continuity. The remaining problem is therefore analytic: a discrete or coarse-grained EABC mechanism must yield a metric estimate strong enough to control the spatial variation of ξ.

Sobolev obstruction in three dimensions. A quadratic lattice energy of Dirichlet type can at best suggest an $H^1 = W^{1,2}$ bound in the continuum. In three dimensions the standard Sobolev embedding gives

$$W^{1,2}(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3), \qquad W^{1,2}(\mathbb{R}^3) \not\hookrightarrow C^{0,\alpha}(\mathbb{R}^3) \quad (\alpha > 0). \tag{17}$$

Consequently, a purely quadratic direction-energy bound does not imply the Hölder control required by geometric depletion criteria for vortex stretching.

Morrey threshold. For $p > 3$, Morrey's inequality yields, locally on regular bounded domains (and in the corresponding whole-space form),

$$W^{1,p} \hookrightarrow C^{0,\alpha}, \qquad \alpha = 1 - \frac{3}{p}. \tag{18}$$

Hence $p = 6$ is the natural sufficient exponent for obtaining the one-half Hölder modulus

$$\boxed{W^{1,6} \hookrightarrow C^{0,1/2}.} \tag{19}$$

Equivalently, for a representative of ξ on a suitable domain Ω,

$$|\xi(x) - \xi(y)| \leq C_\Omega \|\nabla\xi\|_{L^6(\Omega)} |x-y|^{1/2}. \tag{20}$$

This is a sufficient analytic route to a one-half Hölder estimate; it is not claimed to be a necessary or unique route to Navier–Stokes regularity.

Scale-consistent discrete candidate. Let $h > 0$ denote the lattice spacing and let $x \sim y$ denote nearest-neighbour edges of a three-dimensional lattice. The scale-consistent sixth-order finite-difference energy is

$$\boxed{E_h^{(6)}[\xi_h] := h^{-3} \sum_{x\sim y} |\xi_x - \xi_y|^6.} \tag{21}$$

Indeed, $|\xi_x - \xi_y| \sim h|\nabla\xi|$ and the number of edges per unit volume is of order h^{-3}, so that the prefactor $h^{-3} = h^{3-6}$ is the one compatible with a continuum L^6 gradient energy. A uniform estimate

$$\sup_{0<h<h_0} E_h^{(6)}[\xi_h] \le C \tag{22}$$

would therefore be the appropriate discrete compactness input for a $W^{1,6}$-type continuum bound, provided that the interpolation scheme, domain control, and boundary conditions are specified and proved compatible.

A necessary sign-lift refinement of the Fano energy. The Fano norm measures $|\sin\theta|$, not the full vector difference. For unit vectors $a, b \in S^2$,

$$|a-b|^2 = 2(1-a\cdot b), \qquad |a\times b|^2 = (1-a\cdot b)(1+a\cdot b). \tag{23}$$

Thus $|a\times b|$ vanishes for both parallel and antiparallel vectors. It cannot, without an additional orientation condition, control $|a-b|$. This prevents one from inferring (22) directly from a Fano energy of the form

$$E_{h,F}^{(6)} := h^{-3}\sum_{x\sim y} \|\phi(\xi_x,\xi_y,\cdot)\|^6 = h^{-3}\sum_{x\sim y} |\sin\theta_{xy}|^6. \tag{24}$$

A local sign or hemisphere lock is sufficient. For example, if neighbouring directions satisfy

$$\xi_x\cdot\xi_y \ge 0, \tag{25}$$

then

$$|\xi_x-\xi_y|^2 = 2(1-\xi_x\cdot\xi_y) \le 2|\xi_x\times\xi_y|^2, \tag{26}$$

and hence

$$E_h^{(6)}[\xi_h] \le 8E_{h,F}^{(6)}[\xi_h]. \tag{27}$$

Therefore a genuine arithmetic-to-spatial bridge requires two independent ingredients: a uniform sixth-order Fano-angle bound and a mechanism excluding uncontrolled nearest-neighbour sign flips.

Theorem 6.2 (Conditional discrete-to-Morrey bridge)**.** *Let $\Omega \subset \mathbb{R}^3$ be a regular bounded domain and let ξ_h be unit-vector lattice fields admitting a stable interpolation $I_h\xi_h \in W^{1,6}(\Omega)$. Assume that the interpolation satisfies*

$$\|\nabla I_h\xi_h\|_{L^6(\Omega)}^6 \le C_0 E_h^{(6)}[\xi_h] \tag{28}$$

with C_0 independent of h, and that

$$\sup_h E_h^{(6)}[\xi_h] < \infty. \tag{29}$$

Then the interpolants are uniformly bounded in $W^{1,6}(\Omega)$ up to the usual control of their L^6 norm. Morrey's inequality consequently gives

$$[I_h\xi_h]_{C^{0,1/2}(\Omega)} \le C \tag{30}$$

with C independent of h. After extraction of a subsequence one obtains weak $W^{1,6}$ compactness and, for every $\beta < 1/2$, strong compactness in $C^{0,\beta}$ on compact subsets under the standard hypotheses.

The theorem is deliberately conditional: the EABC framework has not yet proved the uniform bound (22), the interpolation estimate, or the hemisphere-lock condition (25). In addition, $\xi = \omega/|\omega|$ is undefined on the zero-vorticity set; a complete Navier–Stokes application must either work on high-vorticity regions or establish a compatible extension procedure.

Relation to geometric depletion. The Constantin–Fefferman direction-of-vorticity program shows that sufficient geometric coherence of vorticity directions can deplete the vortex-stretching nonlinearity. The role of the present subsection is narrower: it identifies a concrete sufficient analytic target by which an EABC lattice mechanism could produce a one-half Hölder modulus. No equivalence with the full hypotheses of any particular classical regularity theorem is asserted here.

Gate status. The resulting status is

$G_{\mathrm{NS}}^{(8a)}$ topological-to-metric implication	FAILED WITHOUT AN ENERGY ESTIMATE,
$G_{\mathrm{NS}}^{(8b)}$ Sobolev obstruction	$W^{1,2}(\mathbb{R}^3) \not\hookrightarrow C^{0,\alpha}$,
$G_{\mathrm{NS}}^{(8c)}$ Morrey threshold	$W^{1,6} \hookrightarrow C^{0,1/2}$ STANDARD THEOREM,
$G_{\mathrm{NS}}^{(8d)}$ uniform discrete $W^{1,6}$ bound	OPEN,
$G_{\mathrm{NS}}^{(8e)}$ Fano-to-vector sign lift	OPEN WITHOUT ORIENTATION LOCK,
$G_{\mathrm{NS}}^{(8f)}$ arithmetic-to-spatial bridge	OPEN,
global 3D Navier–Stokes regularity	NOT ESTABLISHED.

11.6.5 Quartic E_8 Root Isotropy and a Conditional H^2 UV-Coercivity Route

The direct $W^{1,6}$ route of Section 6 asks for a sixth-order discrete gradient bound. A spectrally more natural sufficient route is to obtain quadratic control of two spatial derivatives. In three space dimensions, local H^2 control (together with the standard lower-order localization terms) implies

$$H^2(\Omega) \hookrightarrow W^{1,6}(\Omega) \hookrightarrow C^{0,1/2}(\Omega) \tag{31}$$

on regular bounded domains. Thus an isotropic positive fourth-order Fourier symbol is a natural candidate for a direction-coherence mechanism.

Exact fourth moment of the E_8 root system. Let $R(E_8) \subset \mathbb{R}^8$ denote the standard set of 240 roots, normalized by $|r|^2 = 2$. The normalized root directions form a spherical 7-design. Consequently all spherical moments through degree four coincide with their rotationally invariant spherical averages. In particular, for arbitrary $k \in \mathbb{R}^8$,

Theorem 6.3 (Quartic E_8 root-moment identity)**.**

$$\boxed{\sum_{r\in R(E_8)} (r\cdot k)^4 = 36\,|k|^4.} \tag{32}$$

Proof. Write $u = r/\sqrt{2} \in S^7$. Since the normalized roots are a spherical 7-design, their fourth moment equals the fourth spherical moment. For the unit sphere in dimension $d = 8$,

$$\frac{1}{|S^7|}\int_{S^7} (u\cdot k)^4\, dS(u) = \frac{3}{d(d+2)}|k|^4 = \frac{3}{80}|k|^4. \tag{33}$$

Because $(r\cdot k)^4 = 4(u\cdot k)^4$ and there are 240 roots,

$$\sum_{r\in R(E_8)} (r\cdot k)^4 = 240\cdot 4\cdot \frac{3}{80}|k|^4 = 36|k|^4. \tag{34}$$

□

Equivalently, the fourth moment tensor is

$$\sum_{r\in R(E_8)} r_i r_j r_k r_l = 12\big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big). \tag{35}$$

Equation (32) is therefore an exact algebraic-geometric identity. It is logically independent of the dimensional pole of the E_8 Epstein zeta function at $s = 8$; the latter supplies spectral information about the lattice but does not by itself generate the fourth-order operator below.

Dimensionally consistent three-dimensional pullback. A Navier–Stokes application lives in physical dimension three, whereas the roots r lie in $\mathbb{R}^8$. Therefore the expression $r\cdot\nabla$ is not defined until an embedding is specified. Let

$$\iota : \mathbb{R}^3 \longrightarrow \mathbb{R}^8, \qquad \iota^T\iota = I_3, \tag{36}$$

be an isometric embedding of the physical spatial cotangent space, and define the pulled-back root directions

$$q_r := \iota^T r \in \mathbb{R}^3. \tag{37}$$

For every physical wave vector $\kappa \in \mathbb{R}^3$, applying (32) to $k = \iota\kappa$ gives the exact identity

$$\boxed{\sum_{r\in R(E_8)} (q_r\cdot\kappa)^4 = 36|\kappa|^4.} \tag{38}$$

Thus the quartic isotropy survives every isometric choice of physical three-plane. No preferred three-plane is selected by this identity alone.

Exact quadratic fourth-order form. For a sufficiently regular vector field $\xi : \mathbb{R}^3 \to \mathbb{R}^3$ with suitable decay, define

$$\mathcal{Q}_{E_8}[\xi] := \sum_{r \in R(E_8)} \left\| (q_r \cdot \nabla)^2 \xi \right\|_{L^2}^2. \tag{39}$$

By Plancherel and (38), componentwise in ξ,

$$\boxed{\mathcal{Q}_{E_8}[\xi] = 36 \int_{\mathbb{R}^3} |\kappa|^4 |\widehat{\xi}(\kappa)|^2 \, d\kappa = 36 \, \|D^2 \xi\|_{L^2}^2,} \tag{40}$$

where the last equality uses the standard Frobenius norm of the Hessian. This is an exact isotropic fourth-order quadratic form.

Theorem 6.4 (Conditional E_8 UV-coercivity bridge)**.** *Assume that the effective EABC evolution or action for the vorticity direction contains, on a specified physical three-plane, a positive term*

$$\gamma \, \mathcal{Q}_{E_8}[\xi], \qquad \gamma > 0, \tag{41}$$

and that the resulting a priori energy estimate controls this term uniformly in time together with the lower-order norms required for localization. Then (40) *yields uniform local H^2 control of ξ. On regular bounded domains this implies*

$$\xi \in H^2 \quad \Longrightarrow \quad \xi \in W^{1,6} \quad \Longrightarrow \quad \xi \in C^{0,1/2}, \tag{42}$$

with the corresponding norm bounds.

The theorem is conditional in an essential sense. The exact root identity fixes the isotropic symbol once the quadratic form $\mathcal{Q}_{E_8}$ is present, but it does not prove that the Navier–Stokes evolution contains such a term, does not determine a positive coupling γ, and does not establish an a priori bound for that term. Moreover, since $\xi = \omega/|\omega|$ is undefined on the zero-vorticity set and need not lie in global $L^2(\mathbb{R}^3)$, a complete application must be formulated locally on high-vorticity regions or with a justified extension/localization procedure.

Relation to the spectral pole at $s = 8$. The dimensional pole of the radial E_8 Epstein zeta function at $s = 8$ and the fourth-root moment identity encode different pieces of structure. The pole is a meromorphic spectral invariant tied to dimension and covolume. The identity (32) is a finite root-design identity. A spectral-action derivation connecting the two and producing the positive coefficient γ in (41) is not established here. This separation prevents the pole itself from being interpreted as an unproved source of ultraviolet coercivity.

Gate status. The ninth Navier–Stokes symmetry gate is therefore

$G_{\mathrm{NS}}^{(9a)}$ dimensional E_8 pole at $s = 8$	**EXACT/STANDARD**.
$G_{\mathrm{NS}}^{(9b)}$ quartic root isotropy	$\sum_r (r \cdot k)^4 = 36\|k\|^4$; **EXACT PASS**.
$G_{\mathrm{NS}}^{(9c)}$ 3D isometric pullback	$\sum_r (q_r \cdot \kappa)^4 = 36\|\kappa\|^4$; **EXACT PASS**.
$G_{\mathrm{NS}}^{(9d)}$ positive spectral-action coupling γ	**OPEN**.
$G_{\mathrm{NS}}^{(9e)}$ H^2 coercivity	**CONDITIONAL** on $\gamma > 0$ and a uniform a priori energy bound.
$G_{\mathrm{NS}}^{(9f)}$ $H^2 \to W^{1,6} \to C^{0,1/2}$	**STANDARD ANALYSIS**.
$G_{\mathrm{NS}}^{(9g)}$ vorticity-direction regularity bridge	**CONDITIONAL**.
Global 3D Navier–Stokes regularity	**NOT ESTABLISHED**.